

2024-10-22-Convexity

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Slide 1

Content

Convex Analysis [VJ Chapter 10]

Def 1 $S \subseteq \mathbb{R}^m$ is convex if

for any $z_1, z_2 \in S \implies tz_1 + (1-t)z_2 \in S$

$0 \leq t \leq 1$

Def 2 Convex combination of $z_1, z_2, \dots, z_n \in \mathbb{R}^m$

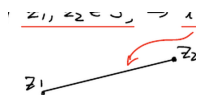
$t_1z_1 + t_2z_2 + \dots + t_nz_n \quad t_i \geq 0$

$\sum_i t_i = 1$

Description

This slide introduces the concept of convex analysis, referencing "VJ Chapter 10." It presents two key definitions: Definition 1 defines a convex set S in $\mathbb{R}^m$ as a set where, for any two points z_1 and z_2 in S , the line segment connecting them (represented by the convex combination $tz_1 + (1-t)z_2$ for $0 \leq t \leq 1$) is entirely contained within S . A figure illustrates this concept with a line segment and an arrow pointing to a point on the line segment. Definition 2 defines a convex combination of points $z_1, z_2, \dots, z_n$ in $\mathbb{R}^m$ as a linear combination $t_1z_1 + t_2z_2 + \dots + t_nz_n$, where each t_i is non-negative and the sum of all t_i equals 1. The slide includes figures illustrating convex combinations in $\mathbb{R}^2$ (a line segment, a triangle) and $\mathbb{R}^3$ (a tetrahedron). The underlining emphasizes the key terms and definitions. The context of linear programming is implicit, as convexity plays a crucial role in the theory and algorithms of linear programming.

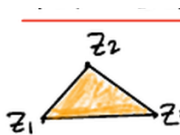
Figures



(a) A line segment connecting two points z_1 and z_2 , with an arrow indicating the convex combination $tz_1 + (1-t)z_2$ lies on the segment.



(b) A line segment connecting two points z_1 and z_2 .



(c) A shaded triangle formed by points z_1 , z_2 , and z_3 .



(d) A shaded tetrahedron formed by points z_1 , z_2 , z_3 , and z_4 .

Slide 2

Content

Thm 10.1 S is convex $\iff$
 it contains all conv. comb. of points in S
pf $\iff$
Suppose S contains all conv. comb. of pts in S
Then clearly, for any $z_1, z_2 \in S$
 $tz_1 + (1-t)z_2 \in S$
 Conv. comb. of z_1, z_2
 $\implies$ S is convex

Description

This slide presents Theorem 10.1, which states that a set S is convex if and only if it contains all convex combinations of its points. The proof is outlined, focusing on one direction of the implication (the reverse direction). It starts by assuming that S contains all convex combinations of its points. Then, it shows that for any two points z_1 and z_2 in S , their convex combination $tz_1 + (1-t)z_2$ is also in S . This directly satisfies the definition of a convex set from the previous slide, thus proving that S is convex. The underlining highlights the key assumptions and conclusions of the proof. The notation "pf" indicates "proof," and the double-headed arrow indicates the "if and only if" nature of the theorem. The term "conv. comb." is an abbreviation for "convex combination." The context of the course (Linear Programming) is crucial for understanding the significance of convex sets in optimization problems.

Figures

Slide 3

Content

Thm 10.1 S is convex $\iff$

it contains all conv. comb. of points in S

pf “ $\implies$ ” Suppose S is convex.

$n = 2$: $z_1, z_2 \in S \implies t_1 z_1 + t_2 z_2 \in S, \quad t_1, t_2 \geq 0$

$t_1 + t_2 = 1$

$n = 3$: $z_1, z_2, z_3 \in S \implies t_1 z_1 + t_2 z_2 + t_3 z_3 = (t_1 + t_2) \left(\frac{t_1}{t_1 + t_2} z_1 + \frac{t_2}{t_1 + t_2} z_2 \right) + t_3 z_3$

$\in S$

$t_1 + t_2 + t_3 = 1$

$t_1 + t_2 \geq 0, \quad t_3 \geq 0$

$\implies t_1 z_1 + t_2 z_2 + t_3 z_3 \in S$

Description

This slide continues the proof of Theorem 10.1, showing that if a set S is convex, then it contains all convex combinations of points in S . This direction of the proof proceeds by induction. The base case, $n = 2$, states that if $z_1, z_2 \in S$, then $t_1 z_1 + t_2 z_2 \in S$ for $t_1, t_2 \geq 0$ and $t_1 + t_2 = 1$. This follows directly from the definition of a convex set. The inductive step, $n = 3$, considers $z_1, z_2, z_3 \in S$. A convex combination $t_1 z_1 + t_2 z_2 + t_3 z_3$ is rewritten as $(t_1 + t_2) \left(\frac{t_1}{t_1 + t_2} z_1 + \frac{t_2}{t_1 + t_2} z_2 \right) + t_3 z_3$, where $t_1 + t_2 + t_3 = 1$ and $t_1, t_2, t_3 \geq 0$.

Since $\frac{t_1}{t_1 + t_2} z_1 + \frac{t_2}{t_1 + t_2} z_2$ is a convex combination of z_1 and z_2 , and S is convex, this term belongs to S . The entire expression is then a convex combination of two elements of S , and thus also belongs to S . This argument can be generalized for any n , completing the proof. The red " $\in S$ " emphasizes that the convex combination of z_1 and z_2 is in S . The arrows indicate the flow of logic in the inductive step. The context of linear programming is relevant because convexity is a crucial property of feasible regions in linear programs.

Figures

Slide 4

Content

Thm 10.1 S is convex $\iff$

it contains all conv. comb. of points in S

pf “ $\implies$ ” Suppose S is convex.

$n = 2$: $z_1, z_2 \in S \implies t_1 z_1 + t_2 z_2 \in S, \quad t_1, t_2 \geq 0$

$t_1 + t_2 = 1$

$n = 4$: $z_1, z_2, z_3, z_4 \in S$

$$t_1 z_1 + t_2 z_2 + t_3 z_3 + t_4 z_4 = (t_1 + t_2 + t_3) \left(\frac{t_1 z_1 + t_2 z_2 + t_3 z_3}{t_1 + t_2 + t_3} \right) + t_4 z_4$$

$(t_1 + t_2 + t_3) + t_4 = 1$

$t_1 + t_2 + t_3 \geq 0, \quad t_4 \geq 0$

$\in S \quad + t_4 z_4 \in S$

Description

This slide continues the proof of Theorem 10.1, specifically the forward direction ("if S is convex, then it contains all convex combinations of points in S "), using induction. It builds upon the $n = 2$ and $n = 3$ cases from the previous slide and now illustrates the case for $n = 4$. It shows how a convex combination of four points, $z_1, z_2, z_3, z_4 \in S$, can be rewritten as $(t_1 + t_2 + t_3) \left(\frac{t_1 z_1 + t_2 z_2 + t_3 z_3}{t_1 + t_2 + t_3} \right) + t_4 z_4$. The term in the parentheses is a convex combination of z_1, z_2 , and z_3 , which belongs to S by the inductive hypothesis (assuming the theorem holds for $n = 3$). The entire expression is then a convex combination of two elements of S (the term in parentheses and z_4), and thus also belongs to S by the definition of convexity. This demonstrates the inductive step, suggesting how the proof can be extended to any n . The red " $\in S$ " emphasizes that the convex combination belongs to S . The arrows indicate how the terms are grouped to form convex combinations. The context of linear programming is relevant because this theorem is fundamental to understanding the properties of feasible regions in linear programs, which are often convex sets.

Figures

Slide 5

Content

Def $S \subseteq \mathbb{R}^m$, Convex Hull of S ,

$$\text{Conv}(S) = \bigcap G$$

G convex, $S \subseteq G$

(= Smallest convex set containing S)

(Note intersection of convex set is convex.)

Thm 10.2 $S \subseteq \mathbb{R}^m$, $\text{Conv}(S)$ = collection of
all conv. comb. of finitely many pts from S

Description

This slide defines the convex hull of a set S in $\mathbb{R}^m$, denoted by $\text{Conv}(S)$. It's defined as the intersection of all convex sets G that contain S . This is equivalent to the smallest convex set containing S . The slide also notes that the intersection of convex sets is convex. Then, Theorem 10.2 is introduced, stating that for a set $S \subseteq \mathbb{R}^m$, the convex hull $\text{Conv}(S)$ is the collection of all convex combinations of finitely many points from S . The underlining emphasizes the definition and the theorem statement. "Conv." is short for "convex," and "comb." is short for "combination." In the context of linear programming, the convex hull is important because it represents the set of all achievable points through convex combinations of points in S , which is relevant for optimization problems.

Figures

Slide 6

Content

Thm 10.2 $S \subseteq \mathbb{R}^m$, $Conv(S)$ = collection of
all conv. comb. of finitely many pts from S

Pf. Let H = collection of all conv. comb.
of finitely many pts from S

Need to show $Conv(S) = H$

(1) $Conv(S) \subseteq H$ ($\because H$ is conv.)

Description

This slide begins the proof of Theorem 10.2, which states that the convex hull of a set S in $\mathbb{R}^m$, denoted $Conv(S)$, is the set of all convex combinations of finitely many points from S . The proof starts by defining H as the set of all convex combinations of finitely many points from S . The goal is to show that $Conv(S) = H$. The first step (1) aims to prove that $Conv(S)$ is a subset of H ($Conv(S) \subseteq H$). The justification for this step relies on the fact that H is convex, indicated by the notation " $\because H$ is conv." The underlining emphasizes what needs to be shown and the first step of the proof. "conv. comb." is short for "convex combination," and "pts" is short for "points." This theorem is important in linear programming because it provides a concrete way to represent the convex hull of a set, which is crucial for understanding the feasible region of linear programs.

Figures

Slide 7

Content

Thm 10.2 $S \subseteq \mathbb{R}^m$, $\text{Conv}(S)$ = collection of
 all conv. comb. of finitely many pts from S
 Pf. Let H = collection of all conv. comb.
 of finitely many pts from S
 Need to show $\text{Conv}(S) = H$
 (2) $H \subseteq \text{Conv}(S)$
 (Thm 10.1, contains all conv. comb.
 of pts from S)

Description

This slide continues the proof of Theorem 10.2. Having established $\text{Conv}(S) \subseteq H$ in the previous slide, this slide focuses on proving the other direction: $H \subseteq \text{Conv}(S)$. It states that this follows from Theorem 10.1, which establishes that a convex set contains all convex combinations of its points. Since $\text{Conv}(S)$ is defined as the smallest convex set containing S , and H is the set of all convex combinations of points from S , Theorem 10.1 implies that $\text{Conv}(S)$ must contain all elements of H . The underlining emphasizes the current step of the proof. "conv. comb." is short for "convex combination," and "pts" is short for "points." This theorem is important in linear programming because it provides a concrete way to represent the convex hull of a set, which is crucial for understanding the feasible region of linear programs.

Figures

Slide 8

Content

Carathéodory Theorem

THEOREM 10.3. The convex hull $\text{conv}(S)$ of a set S in $\mathbb{R}^m$ consists of all convex combinations of $m+1$ points from S :

$$\text{conv}(S) = \left\{ z = \sum_{j=1}^{m+1} t_j z_j : z_j \in S \text{ and } t_j \geq 0 \text{ for all } j, \text{ and } \sum_j t_j = 1 \right\}.$$

$$\text{Pf } \text{Conv}(S) = \left\{ z = \sum_{j=1}^n x_j z_j, \quad z_j \in S, \quad x_j \geq 0, \quad \sum_j x_j = 1 \right\}$$

Consider the LP: $\max C^T X$

s.t. $AX = \underline{z}$

$x_1 + x_2 + \dots + x_n = 1$

$x_1, x_2, \dots, x_n$

(z is given, fixed, $A = [z_1 \quad z_2 \quad \dots \quad z_n]^{(m \times n)}$)

Description

This slide introduces the Carathéodory Theorem (Theorem 10.3). The theorem states that the convex hull of a set S in $\mathbb{R}^m$, denoted $\text{conv}(S)$, can be represented as the set of all convex combinations of at most $m+1$ points from S . The slide provides the formal definition of the convex hull using this representation. The proof begins by restating the definition of the convex hull using n points and introduces a linear program (LP) to aid in the proof. The LP aims to maximize an objective function $C^T X$ subject to the constraints

$AX = z, \sum_{j=1}^n x_j = 1$, and $x_j \geq 0$ for all j . Here, z is a fixed point in the convex hull, and A is a matrix

whose columns are the points z_j from the set S . The notation $[z_1 \quad z_2 \quad \dots \quad z_n]^{(m \times n)}$ indicates that A is an $m \times n$ matrix. The underlining emphasizes the fixed vector z in the LP formulation. In the context of linear programming, Carathéodory's Theorem is important because it shows that any point in the convex hull of a set can be expressed as a convex combination of a relatively small number of points from the set, which has implications for the complexity of algorithms.

Figures

Slide 9

Content

Carathéodory Theorem

THEOREM 10.3. The convex hull $\text{conv}(S)$ of a set S in $\mathbb{R}^m$ consists of all convex combinations of $m + 1$ points from S :

$$\text{conv}(S) = \left\{ z = \sum_{j=1}^{m+1} t_j z_j : z_j \in S \text{ and } t_j \geq 0 \text{ for all } j, \text{ and } \sum_j t_j = 1 \right\}.$$

$$Pf \quad \text{Conv}(S) = \left\{ z = \sum_{j=1}^n x_j z_j, \quad z_j \in S, \quad x_j \geq 0, \quad \sum_j x_j = 1 \right\}$$

Consider the LP: $\max C^T X$

s.t. $AX = \underline{z}$

$x_1 + x_2 + \dots + x_n = 1$

$x_1, x_2, \dots, x_n \geq 0$

$\left. \begin{array}{l} x_1 + x_2 + \dots + x_n = 1 \\ x_1, x_2, \dots, x_n \geq 0 \end{array} \right\} m + 1 \text{ constraints}$

(z is given, fixed, $A = [z_1 \ z_2 \ \dots \ z_n]^{(m \times n)}$)

This is feasible, and hence has a basic opt. soln with $m + 1$ basic variables.

Description

This slide continues the proof of Carathéodory's Theorem (Theorem 10.3). It sets up a linear program (LP) to demonstrate that any point z in the convex hull of a set S in $\mathbb{R}^m$ can be expressed as a convex combination of at most $m + 1$ points from S . The LP is formulated as maximizing a dummy objective function $C^T X$ subject to the constraints $AX = z$, $\sum_{j=1}^n x_j = 1$, and $x_j \geq 0$, where A is a matrix whose columns are the

points z_j from S , and z is the given point in $\text{conv}(S)$. The slide notes that this LP has $m + 1$ constraints. It then states that this LP is feasible, meaning a solution exists, and therefore, it has a basic optimal solution with $m + 1$ basic variables. This implies that at most $m + 1$ of the x_j variables are non-zero, meaning z can be expressed as a convex combination of at most $m + 1$ points from S . The underlining emphasizes the fixed vector z in the LP formulation. "opt. soln" is short for "optimal solution." In the context of linear programming, this theorem is crucial because it provides a bound on the number of points needed to represent any point in the convex hull, which has implications for computational complexity.

Figures